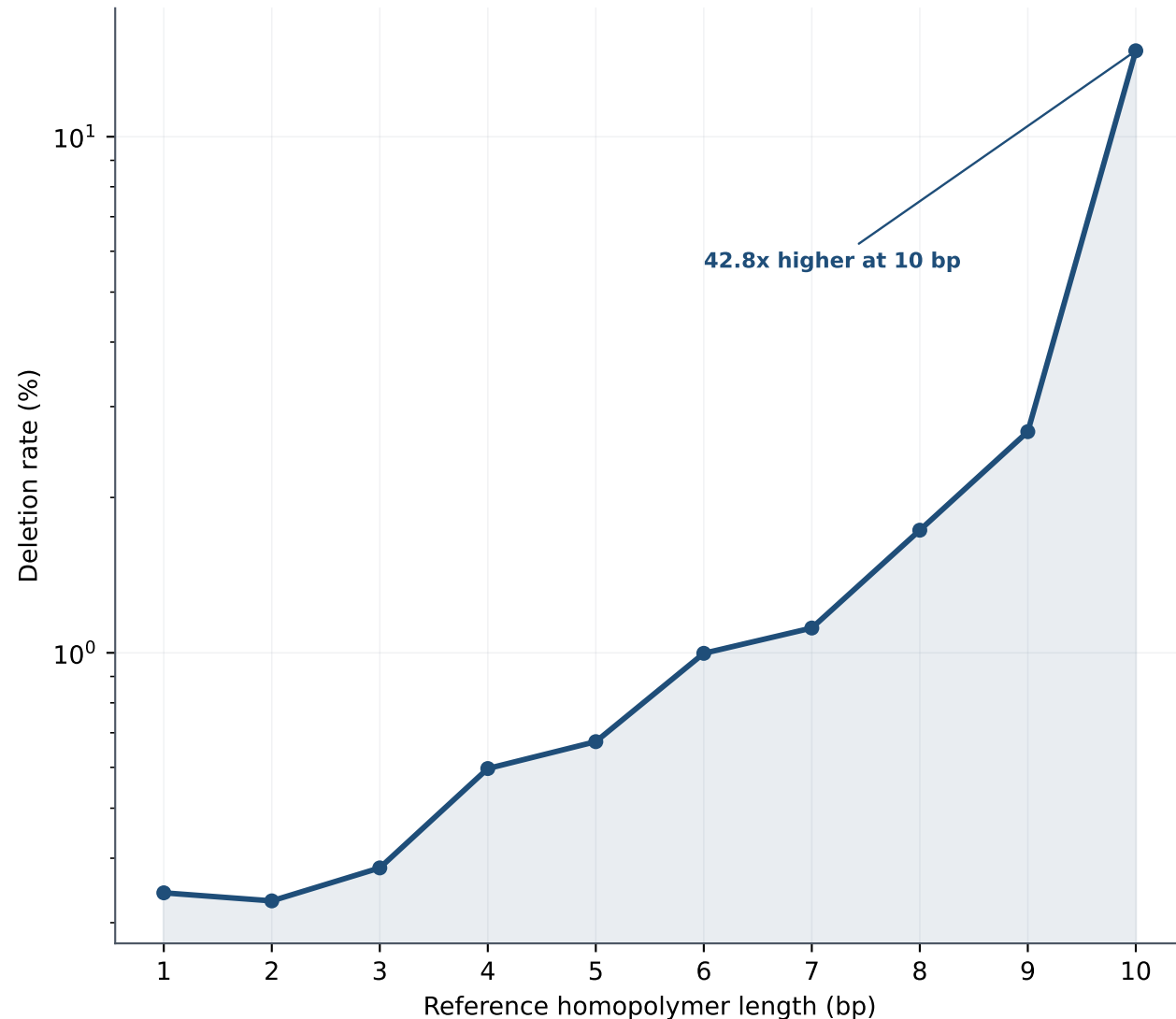


Background Figure B3. Context dependence and the dimensions of recent simulator evaluation

a Real ONT error is context-dependent



b Recent work separates modelling and evaluation layers

Read-level realism

length, accuracy and quality distributions

NanoSim, PBSim3, 2026 benchmark [1,2,16]

Error-structure realism

context-dependent substitutions and homopolymers

2026 benchmark [16]; NanoSimFormer [15]

Signal-level realism

raw signal and basecaller-guided generation

Squigulator, Icarust, transformer models [10-12,15]

Downstream consistency

variant, assembly and metagenomic endpoints

Platinum Pedigree, NanoSimFormer [13,15]

Instrument-claim alignment

the metric must contain the property being claimed

GCerrHMM, this study